RAJALAKSHMI ENGINEERING COLLEGE

Deepika D P 251801049-2025-AIDS

D2

Started on	Thursday, 17 September 2026, 11:18 AM
State	Finished
Completed on	Thursday, 17 September 2026, 11:33 AM
Time taken	15 mins 13 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format
First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
long long countways(int n){
    if(n==0||n==1||n==2){
        return 1;
    }
    long long w0 = 1;
    long long w1 = 1;
    long long w2 = 1;
    long long current_w = 0;
    for(int i=3;i<=n;i++){
        current_w = w2+w0;
        w0=w1;
        w1=w2;
        w2=current_w;
    }
    return w2;
}
int main(){
    int n;
    if(scanf("%d",&n)!=1){
        return 1;
    }
    printf("%lld\n",countways(n));
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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